Meet 0

October 19, 2026

Team

You have 30 minutes to complete 6 problems. There is no penalty for guessing. Answers are only valid if they are written in the below table.

First and last name			
p	Answer	p	Answer
1		4	
2		5	
3		6	

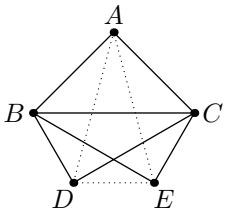
1. Suppose p is a prime such that $q = (p^2 - 1)^2 + 3$ is also a prime. Compute q.

2. Compute the sum of the numerator and denominator of the expression

$$1 + \frac{1}{2 + \frac{1}{4 + \frac{1}{8}}}$$

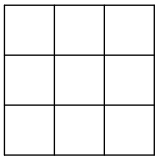
when fully simplified.

3. In $\triangle ABC$, $\overline{AB} = \overline{AC}$, $\overline{BC} = 10$ and $\angle A = 90^\circ$. Point D lies on the opposite side of $\overline{BC}$ as A, such that $\angle DBC = 60^\circ$ and $\angle DCB = 30^\circ$. Point E lies on the same side of $\overline{BC}$ as D such that $\angle EBC = 30^\circ$ and $\angle ECB = 60^\circ$. The area of $\triangle ADE$ is $\frac{a+b\sqrt{c}}{d}$ where a, b, c, and d are positive integers with $\gcd(a, b, d) = 1$ and c is not divisible by the square of any prime. Compute $a + b + c + d$.



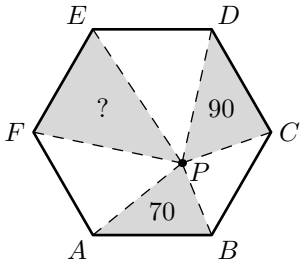
4. Place digits in a 3×3 grid such that

   - when the rows are read from left to right, the first row is a multiple of 67, the second row is $n^2 + 2n$ for an integer n, the third row is $a^b + b^a$ for integers $a, b > 1$,
   - when the columns are read from top to bottom, the first column is a perfect square, the second column is a Fibonacci number, and when the third column is multiplied by 7, the last three digits are 999,
   - and none of the rows or columns start with a 0.



   Compute the sum of the three-digit numbers formed by the rows.

5. Regular hexagon $ABCDEF$ has area 600. Point P lies inside the hexagon so that $\triangle PAB$ has area 70 and $\triangle PCD$ has area 90. Find the area of $\triangle PEF$.



6. Michael owns 10 000 dollars in Minnetonka Math Team stock. Each day, the stock randomly goes either up or down ten percent with equal probability. After four days, Michael's stock is worth d dollars. Compute the expected value of $(d + 6)(d + 7)$.